

Initial Post:

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by [Abdulla Husain Salem Hadna Almessabi](#) - Thursday, 7 August 2025, 1:13 PM

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Initial Post: Impact of Industry 4.0 and Industry 5.0 on the Manufacturing Sector

The shift of Industry 4.0 to Industry 5.0 has resulted in some major modifications in the manufacturing industry. Cyber-physical systems, automation, and IoT have transformed production efficiency and data exchange (Aljaidi et al., 2022). Industry 5.0 changed the facet of human-centric processes that prioritize the health of employees and sustainability (Vatsyayan et al., 2022).

One of the real-life examples of the consequences of failure in an information system is the WannaCry ransomware attack on Nissan in 2017. Nissan information systems became the target of the attack, which led to the shutdown of some of the company's operations in the UK and globally (Reinhold, 2024). Ransomware kept the company at a standstill and loss of revenue (Farion-Melnyk et al., 2021). It impacted manufacturing and logistical activities, and it caused the delay of thousands of vehicle deliveries.

This was costly because Nissan suffered a high revenue loss as a result of production and logistic stoppage. In addition to the financial one, there was a harsh reputational cost as well. Partners and customers were behind the party, raising questions related to the cybersecurity measures implemented at Nissan and weakening the reputation of the company in terms of resistance to cyberattacks due to the increased dependence on IoT and automation, as part the Industry 4.0 (Nwankpa and Schooley, 2025).

This event highlights the difficulties of protecting networks of systems in an online era. As we move to Industry 5.0, we need to look to human-focused approaches to solve such weaknesses. The companies can reduce potential risks connected to cyberattacks by paying more attention to enhancing cybersecurity protocols and adding human control over the automated work processes, and guarantee a more secure and sustainable future.

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